METHOD IN MOTION

# THINK LIKE A PROGRAMMER

VOLUME 1: CODING FOUNDATIONS

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Foundational Computational Thinking Concepts for Future Programmers

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METHOD IN MOTION

# WELCOME FUTURE PROGRAMMER

Programming is more than typing code into a computer. Great programmers learn how to think critically, solve problems, recognize patterns, and follow logical steps long before they write their first line of code.

This guide introduces foundational computational thinking concepts through simple explanations, real-life examples, and beginner-friendly exercises designed to help future programmers build strong problem-solving skills.

## INSIDE THIS VOLUME

- // Sequencing
- // Patterns
- // Logic
- // Algorithms
- // Debugging
- // Variables
- // Loops
- // Problem Solving
- // And more

*"Every expert programmer starts with the fundamentals."*

# SEQUENCING

## WHAT IS IT?

Sequencing means putting steps in the right order. Coding uses sequencing because computers follow instructions one step at a time. If the steps are mixed up, the computer may not do what you wanted. Kids use sequencing every day when they get dressed, brush their teeth, or clean up toys. Learning sequencing helps children think clearly, follow directions, and explain ideas in order.

## KEY TERMS

Sequence, Steps, Order, Instructions

## REAL LIFE USE

Making a sandwich uses sequencing because each step has to happen in the right order.

## TRY THIS

List 3 things you do every morning in order.

## QUICK TIP

Computers follow instructions in order.

# PATTERNS

## WHAT IS IT?

Patterns are things that repeat in a way we can recognize. Patterns can be colors, sounds, shapes, numbers, movements, or routines. Coders use patterns to understand problems and make programs work more smoothly. When kids notice patterns, they build prediction skills and stronger logical thinking. Pattern practice also helps children see how repeated actions connect to coding.

## KEY TERMS

Pattern, Repeat, Predict, Routine

## REAL LIFE USE

A clapping game uses a pattern when the same beat repeats again and again.

## TRY THIS

Find a pattern somewhere in your home.

## QUICK TIP

Patterns help coders make predictions.

# LOGIC

## WHAT IS IT?

Logic means thinking carefully to make a choice or solve a problem. Coders use logic to decide what should happen next in a program. Kids use logic when they follow rules, make decisions, or figure out why something happened. Learning logic helps children slow down, think step by step, and explain their reasoning clearly.

## KEY TERMS

Logic, Decision, Reasoning, Problem Solving

## REAL LIFE USE

If it is raining, you bring an umbrella. That is logic.

## TRY THIS

Name one choice you made today and why you made it.

## QUICK TIP

Logic helps computers make decisions.

# ALGORITHMS

## WHAT IS IT?

An algorithm is a set of steps used to finish a task. Computers use algorithms to follow directions and solve problems. Kids use algorithms when they follow recipes, build with blocks, or play games with rules. Learning algorithms helps children understand that big tasks can be broken into smaller steps.

## KEY TERMS

Algorithm, Steps, Task, Rules

## REAL LIFE USE

A recipe is an algorithm because it gives steps to follow.

## TRY THIS

Write the steps for making your favorite snack.

## QUICK TIP

Algorithms turn big tasks into small steps.

# DEBUGGING

## WHAT IS IT?

Debugging means finding and fixing mistakes. Coders debug when something does not work the way they expected. Kids debug when they fix a puzzle, correct schoolwork, or figure out why a toy is not working. Debugging teaches patience, problem solving, and confidence. Mistakes are part of learning.

## KEY TERMS

Debugging, Mistake, Fix, Problem

## REAL LIFE USE

If a toy will not turn on, checking the batteries is debugging.

## TRY THIS

Think of one mistake you fixed today.

## QUICK TIP

Debugging means learning from mistakes.

# COMMANDS

## WHAT IS IT?

A command is an instruction that tells someone or something to do one action. In coding, commands tell the computer what to do. Screen-free commands can be things like clap, jump, turn, stop, or repeat. Learning commands helps kids understand that coding is made from small actions.

## KEY TERMS

Command, Action, Instruction, Code

## REAL LIFE USE

"Jump three times" is a command someone can follow.

## TRY THIS

Give a command to a friend or family member.

## QUICK TIP

Commands tell computers what to do.

# LOOPS

## WHAT IS IT?

A loop means repeating something again and again. Coders use loops so they do not have to write the same instruction many times. Kids use loops in songs, dances, games, and routines. Learning loops helps children understand repetition and efficiency.

## KEY TERMS

Loop, Repeat, Again, Cycle

## REAL LIFE USE

Singing the same chorus in a song is like a loop.

## TRY THIS

Find something you repeat every day.

## QUICK TIP

Loops help coders save time.

# CONDITIONALS

## WHAT IS IT?

A conditional is an if/then idea. It means something happens only if something else is true. Coders use conditionals to help computers make decisions. Kids use conditionals every day, like if it is cold, then wear a jacket. Learning conditionals helps children connect choices with results.

## KEY TERMS

Conditional, If Then, Choice, Result

## REAL LIFE USE

If your hands are dirty, then you wash them.

## TRY THIS

Make your own if/then sentence.

## QUICK TIP

Conditionals help programs make choices.

# VARIABLES

## WHAT IS IT?

A variable is like a container that holds information. In coding, variables can hold numbers, names, scores, or answers. Kids can understand variables by thinking about labels, boxes, or containers. Learning variables helps children understand how information can be stored and changed.

## KEY TERMS

Variable, Information, Store, Change

## REAL LIFE USE

A backpack can hold different things, just like a variable can hold information.

## TRY THIS

Name something that can change, like age or score.

## QUICK TIP

Variables hold information.

# SORTING

## WHAT IS IT?

Sorting means putting things into groups. Coders sort information to make it easier to find, use, or understand. Kids sort toys, clothes, colors, shapes, and snacks all the time. Sorting builds organization skills and helps children notice similarities and differences.

## KEY TERMS

Sorting, Groups, Same, Different

## REAL LIFE USE

Sorting laundry by color is a real-life sorting task.

## TRY THIS

Sort 5 objects into two groups.

## QUICK TIP

Sorting makes information easier to use.

# DECOMPOSITION

## WHAT IS IT?

Decomposition means breaking a big problem into smaller parts. Coders use decomposition when a project feels too big to solve at once. Kids use it when cleaning a room, building a toy, or planning a project. This skill helps children feel less overwhelmed and more in control.

## KEY TERMS

Decomposition, Break Down, Parts, Problem

## REAL LIFE USE

Cleaning a room is easier when you split it into smaller tasks.

## TRY THIS

Break one big chore into 3 smaller steps.

## QUICK TIP

Big problems are easier in small parts.

# CAUSE AND EFFECT

## WHAT IS IT?

Cause and effect means one thing makes another thing happen. Coders use cause and effect to understand how actions create results. Kids see cause and effect when they press a button, knock over blocks, or follow rules in a game. This builds strong thinking skills for coding.

## KEY TERMS

Cause, Effect, Action, Result

## REAL LIFE USE

If you push a toy car, it moves forward. That is cause and effect.

## TRY THIS

Name one action and the result it creates.

## QUICK TIP

Code works through cause and effect.

# PROBLEM SOLVING

## WHAT IS IT?

Problem solving means figuring out what to do when something is hard or confusing. Coders solve problems constantly by testing ideas and trying again. Kids problem solve when they build, play, share, or fix mistakes. Learning problem solving helps children become patient, creative, and confident thinkers.

## KEY TERMS

Problem, Solution, Try, Think

## REAL LIFE USE

Building a tall block tower takes problem solving when it keeps falling.

## TRY THIS

Solve one small problem without asking for help first.

## QUICK TIP

Coders keep trying until they find a solution.

# TESTING

## WHAT IS IT?

Testing means trying something to see if it works. Coders test their programs to make sure the instructions do what they should. Kids test ideas when they build, create, cook, or play. Testing helps children learn that it is okay to try, check, and improve.

## KEY TERMS

Testing, Try, Check, Improve

## REAL LIFE USE

Trying a puzzle piece to see if it fits is testing.

## TRY THIS

Test one idea and see what happens.

## QUICK TIP

Testing helps coders improve their work.

VOLUME 1 SUMMARY

# THINK LIKE A PROGRAMMER

## CODING FOUNDATIONS

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In this volume, you explored the essential building blocks of programming before writing code. You learned how sequencing, patterns, logic, algorithms, and problem solving help you think like a coder and create solutions.

You also discovered how loops, conditionals, variables, debugging, and commands help computers follow instructions, make decisions, repeat actions, and fix mistakes. These foundational concepts are used in every program, app, game, website, and digital system.

Strong foundations build confident problem solvers and prepare you for more advanced programming in the next volumes.

## KEY DEFINITIONS

- **Sequencing** — Putting steps in the right order to complete a task.
- **Patterns** — Recognizing things that repeat to make predictions.
- **Logic** — Using reasoning to make decisions and solve problems.
- **Algorithms** — Step-by-step plans used to finish tasks and solve problems.
- **Debugging** — Finding and fixing mistakes in code or instructions.
- **Commands** — Instructions that tell a computer or person to do an action.
- **Loops** — Repeating actions multiple times to save time and effort.
- **Conditionals** — If/then decisions that control what happens next.
- **Variables** — Containers that store information that can change.
- **Problem Solving** — Breaking challenges down and finding the best solutions.
- **Critical Thinking** — Analyzing situations and choosing the best approach.
- **Computational Thinking** — Using logic and structured steps to solve problems.

*"Strong foundations today build extraordinary creations tomorrow."*